

Effect of urine pH changed by dietary intervention on uric acid clearance

mechanism of pH-dependent excretion of urinary uric acid

Background The finding reported in a previous paper - alkalization of urine facilitates uric acid excretion - is contradictory to what one might expect to occur: because food materials for the alkalization of urine contain fewer purine bodies than those for acidification, less uric acid in alkaline urine should have been excreted than in acid urine. To make clear what component of uric acid excretion mechanisms is responsible for this unexpected finding, we simultaneously collected data for the concentration of both creatinine and uric acid in serum as well as in urine, in order to calculate both uric acid and creatinine clearances. **Methods** Within the framework of the Japanese government's health promotion program, we made recipes which consisted of protein-rich and less vegetable-fruit food materials for H⁺-load (acidic diet) and others composed of less protein and more vegetable-fruit rich food materials (alkaline diet). This is a crossover study within some limitations. Healthy female students, who had no medical problems at the regular physical examination provided by the university, were enrolled in this consecutive 5-day study for each test. From whole-day collected urine, total volume, pH, organic acid, creatinine, uric acid, titratable acid and all cations (Na⁺, K⁺, Ca²⁺, Mg²⁺, NH₄⁺) and anions (Cl⁻, SO₄²⁻, PO₄⁻) necessary for the estimation of acid-base balance were measured. In the early morning before breakfast of the 1st, 3rd and 5th experimental day, we sampled 5 mL of blood to estimate the creatinine and uric acid concentration in serum. **Results and discussion** Urine pH reached a steady state 3 days after switching from ordinary daily diets to specified regimens. The amount of acid generated ([SO₄²⁻] + organic acid – gut alkali) was linearly related with the excretion of acid (titratable acid + [NH₄⁺] – [HCO₃⁻]), indicating that H⁺ in urine is generated by the metabolic degradation of food materials. Uric acid and excreted urine pH retained a linear relationship, as reported previously. Among the five factors which are associated with calculating clearances for both uric acid and creatinine, we identified a conspicuous difference between acidic and alkaline diets in the uric acid concentration in serum as well as in urine; uric acid in the serum was higher in the acidic group than in the alkaline group, while uric acid in the urine in the acidic group was lower than that in the alkaline group. These changes of uric acid in acidic urine and in serum were reflected in the reduction of its clearance. From these observations, it is considered that uric acid may be reabsorbed more actively in acidic urine than in alkaline urine. **Conclusion** We conclude that alkalization of urine by eating nutritionally well-designed alkaline-prone food is effective for removing uric acid from the body.